Chemical changes

Predict reactions

1 Reactivity & displacement

More reactive metals form positive ions more readily and displace less reactive metals.

Mg > Zn > Fe > H > Cu: Mg/Zn/Fe react with dilute HCl or H_2SO_4; Cu does not.

$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$.

2 Redox & extraction

Oxidation: gain oxygen or lose electrons.
Reduction: lose oxygen or gain electrons.

Below carbon: oxide can be reduced using carbon. More reactive metals need electrolysis. Gold may occur as the element.

Make & measure salts

3 Acids & salts

$H^+ + OH^- \rightarrow H_2O$. Metal + acid → salt + H_2; oxide/hydroxide + acid → salt + water.

Carbonate + acid also gives CO_2. HCl → chlorides; H_2SO_4 → sulfates; HNO_3 → nitrates.

Strong = fully ionised; concentrated = much solute per volume.

4 Three practical routes

Insoluble base + acid: add excess, filter, concentrate, cool, filter/dry crystals.

Titration: pipette alkali, burette acid, indicator; swirl, dropwise endpoint, concordant titres.

Electrolysis: inert electrodes, DC; predict and test separate electrode products.

Drive electrolysis

5 Electrode rules

Molten/dissolved ions can move. Positive ions → negative cathode → reduction. Negative ions → positive anode → oxidation.

Molten binary compound: metal at cathode, non-metal at anode.

Aqueous: H_2 instead of metals above H; halogen at anode if halide, otherwise O_2.

6 Aluminium & half-equations

Cryolite lowers melting/operating temperature. $Al^{3+} + 3e^- \rightarrow Al$.

$2O^{2-} \rightarrow O_2 + 4e^-$. Oxygen reacts with carbon anodes, so replace them.

Balance atoms AND charge. Electrons travel in wires; ions carry current through electrolyte.